

Math 110 – Homework 5 Answers

Math 110 – Fall 2026

Homework 5

Quotient spaces, isomorphism theorems, and duality

Suggested completion date: Friday, October 2

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Define

$$T : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathbb{R}^2, \quad T(p) = (p(0), p(1)).$$

(a) Determine $\text{null } T$ and prove that T is surjective.

(b) Describe explicitly the induced isomorphism

$$\mathcal{P}_3(\mathbb{R}) / \text{null } T \longrightarrow \mathbb{R}^2$$

supplied by the first isomorphism theorem.

(c) Find a basis of $\mathcal{P}_3(\mathbb{R}) / \text{null } T$.

Problem 2. Let $T : V \rightarrow W$ be linear and let U be a subspace of V contained in $\text{null } T$. Prove that there is a unique linear map

$$\tilde{T} : V/U \longrightarrow W$$

such that $\tilde{T}(v + U) = Tv$ for every $v \in V$. Determine $\text{null } \tilde{T}$ in terms of $\text{null } T$ and U .

Problem 3. (LADR §3F, Exercise 1.) Prove that every linear functional $\varphi : V \rightarrow \mathbb{F}$ is either the zero map or is surjective.

Problem 4. Let $v_1, \dots, v_n$ be a basis of a finite-dimensional vector space V , and let $\varphi_1, \dots, \varphi_n$ be its dual basis.

(a) Prove that every $v \in V$ satisfies

$$v = \sum_{j=1}^n \varphi_j(v) v_j.$$

(b) If $\psi \in V'$, prove that

$$\psi = \sum_{j=1}^n \psi(v_j) \varphi_j.$$

Problem 5. Let $T : V \rightarrow W$ be linear between finite-dimensional spaces. If A is the matrix of T with respect to bases $\mathcal{B}$ of V and $\mathcal{C}$ of W , prove that the matrix of the dual map

$$T' : W' \longrightarrow V', \quad T'(\psi) = \psi \circ T,$$

with respect to the corresponding dual bases is A^T .

Problem 6. Let

$$U = \text{span}((1, 1, 0), (0, 1, 1)) \subseteq \mathbb{R}^3.$$

(a) Determine the annihilator $U^0 \subseteq (\mathbb{R}^3)'$, identifying a functional with its row vector.

(b) Verify in this example that $\dim U + \dim U^0 = 3$.

Problem 7 (optional). Let $j_V : V \rightarrow V''$ be defined by $j_V(v)(\varphi) = \varphi(v)$. Prove that j_V is injective, and prove that it is an isomorphism when V is finite-dimensional.

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5

Solution notes

Problem 6

Solution notes

Problem 7

Solution notes
